Variation of parameters with the Wronskian

We've already seen how to use variation of parameters with a system of linear equations to find the solution to a second order nonhomogeneous equation.

When we looked at this method before, we built this system of linear equations:

$$u_1'y_1 + u_2'y_2 = 0$$

$$u_1'y_1' + u_2'y_2' = g(x)$$

If we solve the first equation for u_1',

$$u_1' = -\frac{u_2'y_2}{y_1}$$

then we can substitute it into the second equation to solve it for u_2'.

$$\left(-\frac{u_2'y_2}{y_1}\right)y_1' + u_2'y_2' = g(x)$$

$$u_2'\left(y_2' - \frac{y_2y_1'}{y_1}\right) = g(x)$$

$$u_2'\left(\frac{y_1y_2' - y_2y_1'}{y_1}\right) = g(x)$$

$$u_2' = \frac{y_1g(x)}{y_1y_2' - y_2y_1'}$$

© Krista King Math

Plugging this back into the equation we found for u_1', we get

$$u_1' = -\frac{y_1 y_2 g(x)}{y_1(y_1 y_2' - y_2 y_1')}$$

$$u_1' = -\frac{y_2 g(x)}{y_1 y_2' - y_2 y_1'}$$

Notice that the denominators of u_1' and u_2' are equivalent, they're both $y_1 y_2' - y_2 y_1'$, and this value is actually the Wronskian of the fundamental set of solutions $\{y_1, y_2\}$.

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_2 y_1'$$

We know that the Wronskian of a fundamental set of solutions is non-zero, which means that the denominators of u_1' and u_2' won't be 0, which means they'll be defined.

Now to find the values of u_1 and u_2, we need to integrate u_1' and u_2'.

$$u_1 = \int u_1' \, dx = \int -\frac{y_2 g(x)}{y_1 y_2' - y_2 y_1'} \, dx = \int -\frac{y_2 g(x)}{W(y_1, y_2)} \, dx$$

$$u_2 = \int u_2' \, dx = \int \frac{y_1 g(x)}{y_1 y_2' - y_2 y_1'} \, dx = \int \frac{y_1 g(x)}{W(y_1, y_2)} \, dx$$

Given these values of u_1 and u_2, using variation of parameters and these Wronskian integrals tells us that the particular solution to the second order nonhomogeneous equation will be

$$y_p(x) = u_1 y_1 + u_2 y_2$$

© Krista King Math

$$y_p(x) = -y_1 \int \frac{y_2 g(x)}{W(y_1, y_2)}\, dx + y_2 \int \frac{y_1 g(x)}{W(y_1, y_2)}\, dx$$

Then, just as before, to find the general solution to the second order nonhomogeneous equation, we add this particular solution to the complementary solution.

$$y(x) = y_c(x) + y_p(x)$$

Before we look at an example, realize that this Wronskian integral method for variation of parameters is the same as the system of equations method for variation of parameters that we used earlier. These are just two different ways of presenting an identical process, so we can use either method.

That being said, let's rework the example we solved previously with a system of equations for variation of parameters, but this time we'll use the Wronskian integrals.

Example

Use variation of parameters to find the general solution to the differential equation.

$$y'' + 4y' + 4y = \frac{e^{-2x}}{x^3}$$

© Krista King Math



Just as before, we solve the characteristic equation associated with the homogeneous equation to find equal real roots and a complementary solution.

$$y_c(x) = c_1 e^{-2x} + c_2 x e^{-2x}$$

Then the fundamental set of solutions and their derivatives are

$$\{y_1, y_2\} = \{e^{-2x}, xe^{-2x}\}$$

$$\{y_1', y_2'\} = \{-2e^{-2x}, e^{-2x} - 2xe^{-2x}\}$$

The Wronskian of this fundamental set is

$$W(e^{-2x}, xe^{-2x}) = \begin{vmatrix} e^{-2x} & xe^{-2x} \\ -2e^{-2x} & e^{-2x} - 2xe^{-2x} \end{vmatrix}$$

$$W(e^{-2x}, xe^{-2x}) = (e^{-2x})(e^{-2x} - 2xe^{-2x}) - (xe^{-2x})(-2e^{-2x})$$

$$W(e^{-2x}, xe^{-2x}) = e^{-4x} - 2xe^{-4x} + 2xe^{-4x}$$

$$W(e^{-2x}, xe^{-2x}) = e^{-4x}$$

Using the Wronskian integrals, the particular solution is

$$y_p(x) = -e^{-2x} \int \frac{xe^{-2x}\left(\frac{e^{-2x}}{x^3}\right)}{e^{-4x}}\, dx + xe^{-2x} \int \frac{e^{-2x}\left(\frac{e^{-2x}}{x^3}\right)}{e^{-4x}}\, dx$$

$$y_p(x) = -e^{-2x} \int \frac{1}{x^2}\, dx + xe^{-2x} \int \frac{1}{x^3}\, dx$$

$$y_p(x) = -e^{-2x}\left(-\frac{1}{x}\right) - \frac{1}{2} xe^{-2x} x^{-2}$$

© Krista King Math

$$y_p(x) = \frac{2e^{-2x}}{2x} - \frac{e^{-2x}}{2x}$$

$$y_p(x) = \frac{e^{-2x}}{2x}$$

This is the same particular solution we found in the earlier lesson when we used a system of equations with variation of parameters. So, just as before, adding this particular solution to the complementary solution gives us the general solution $y(x)$.

$$y(x) = c_1 e^{-2x} + c_2 x e^{-2x} + \frac{e^{-2x}}{2x}$$

© Krista King Math